Grandhi Lakshmi Narayana Guptha

ee25mtech11030@iith.ac.in | gupthagrandhi96@gmail.com | +91 9704677490 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.08
Sri chaitnaya junior college	Class XII (APBIE)	2021	93.90%
Dr KKR's gowtham school	Class X (CBSE)	2019	90.40%

PROJECTS

Runtime SOC/SOH Estimation for Li-ion Batteries from ECM Parameters - (JULY - 2026)

- Built ECM parameter-extraction pipeline (OCV, R0, R1, C1, SOH) from PulseBat HPPC data across 203 batteries

(LMO/NMC/LFP) and real CALCE hardware

- Designed filter-free dual-forgetting-factor RLS (FFRLS) runtime estimator with trust-region clipping and no Coulomb counting
- Achieved 1.74-1.99% SOC error (synthetic cycles) and 3.35-4.13% SOC error (real CALCE hardware, 4 drive cycles)

Closed-Loop BLDC Motor Drive Simulation - (MAY - 2026)

- Modelled 2 kW, 48 V, 3000 rpm, 8-pole BLDC motor from commercial datasheet parameters
- Designed cascaded PI speed/current control loop for closed-loop operation
- Implemented and cross-validated control design on dual PLECS and MATLAB/Simulink platforms

Wireless Data & Power Transfer System - (Apr-2026)

- Built two-node Arduino Mega wireless system using LoRa SX1278 modules
- Designed custom MIL-STD-1553-inspired communication protocol for the LoRa link
- Integrated inductive wireless power transfer alongside the data link for combined data+power delivery

Closed-Loop DC-DC Boost Converter -- Hardware Development & PCB Design - (APR - 2026)

- Developed closed-loop DC-DC boost converter hardware end-to-end for Power Electronics Lab
- Designed 200 W boost-converter inductor for 48-62 V in / 64-85 V out at 80 kHz, 95% efficiency target
- Designed and routed the fabrication-ready PCB (KiCad 9.0.8) the converter hardware was built on
- Implemented digital closed-loop control on TMS320F28049C (C2000) interfaced to the PCB

Prime Number Checker Using Verilog - (MAY-2024)

- Designed prime-number-checking digital logic circuit using a Finite State Machine (FSM) in System Verilog HDL
- Implemented evaluation logic against the prime-number condition for arbitrary inputs
- Synthesized and implemented the design on the Basys 3 FPGA development board (Xilinx Artix-7 XC7A35T) using Vivado.

WORK EXPERIENCE

R&D Inten at Qbit Labs pvt ltd - (JAN-2025):

- Developed/modified Verilog/SystemVerilog RTL for Rx Coarse Sync chain of HAL modem Advanced PHY (Vivado)
- Worked on modem implementations supporting different MSPS configurations.
- Generated FPGA bitfiles and validated designs on hardware using Python-automated board-level testing

Electrical Maintenance Intern

- **Supported motor & transformer maintenance** through dismantling, rewinding, insulation treatment and varnishing, gaining hands-on exposure to industrial electrical-machine repair.
- **Verified repaired-machine electrical integrity** using Megger, LCR meter and HV/surge testing, ensuring insulation,

winding and inter-turn fault-free conditions.
- **Assessed post-repair machine performance** through no-load testing and vibration, noise and temperature monitoring, supporting reliable machine operation.

SKILLS

Software : MATLAB, PLECS, Ki Cad, NI Multisim, Vivado.

programming language : Matlab, Python, HTML.

Hardware programming language : System Verilog , Verilog.

RELEVANT COURSES

- Multi Level Converters
- Permanent Magnet Synchronous Machines
- Electrical machines Analysis and Control
- Embedded System Design in Power Electronics

POSITIONS OF RESPONSIBILITY

Class Representative - (2023-2025)

- B.Tech Class Representative, facilitating smooth communication and relation between the student body and faculty.

SCHOLASTIC ACHIEVEMENTS

Tutor and Expert - (2023-2025)

- solved problems as a Tutor in Tutorbin and also an Expert in Chegg

EXTRACURRICULAR

Gold Medalist - Cricket -2026

- Member of the gold-medal-winning team in the inter-hostel cricket competition at IIT Hyderabad's annual General Championship.

ICPS - 2025

- Managed registration desk operations at the 11th International Conference on Power Systems (ICPS), coordinating check-in and credentialing for 400+ participants.